RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
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Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING
(Common to CS, IS & AI)

Date: 17/12/2025	CIE - II	Max. Marks: 60
Semester: V	UG	Duration: 2 Hour
Course Title: ADVANCED ALGORITHMS		Course Code: CS355TBB

Sl. No.	PART A	M	BT	CO
1.	A residual graph is _____	1	1	1
2.	The time complexity of Ford–Fulkerson algorithm can be given as _____	1	2	2
3.	Differentiate between maximum matching and maximum cardinality matching.	2	2	3
4.	Number of iterations in Bellman–Ford for V vertices is _____	1	2	1
5.	Bipartite matching can be solved using _____	1	1	2
6.	Give two real-life applications of greedy algorithms.	2	3	3
7.	Give two reasons that make Johnson's algorithm well-suited for handling sparse graphs.	2	3	2
	PART B			
1a.	Given the following directed graph, compute shortest distances from source S using Bellman–Ford and detect if a negative cycle exists: Edges: S → A (4), S → B (5), A → C (5), B → C (–3) and C → B (2)	6	3	1
1b.	Explain why DAG shortest path algorithms run faster than Dijkstra or Bellman–Ford algorithms with an example.	4	3	2
2.	Discuss the Matrix-Chain Multiplication problem using Dynamic Programming in detail.	10	3	4
3.	Consider the flow network below: Edges with capacities: S → A (10), S → C (10), A → B (4), A → C (2), C → D (9), D → B (6), B → T (10) and D → T (10) Compute the maximum flow and show the residual graph after each iteration.	10	2	2
4a.	Given the DAG with the below edges: A → B (2), A → C (5), B → D (1), C → D (2) and D → E (3) Perform topological sorting and also compute shortest path from A to all other vertices.	6	3	1
4b.	Describe the conditions under which the Ford–Fulkerson algorithm may fail to terminate when dealing with irrational edge capacities.	4	2	3

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5.	Consider the below activities with start and finish times: **Activity** **Start** **Finish** 1 1 4 2 3 5 3 0 6 4 5 7 5 3 9 6 5 9 7 6 10 8 8 11 9 8 12 Select the maximum set of non-overlapping activities using the greedy strategy. Provide justification for the same.	10	3	1

Course Outcomes: After completing the course, the students will be able to: -	
CO 1	Analyze various algorithms for their time and space complexity.
CO 2	Demonstrate a familiarity with major algorithms and data structures
CO 3	Apply appropriate design techniques for solving real world problems.
CO 4	Design and implement solutions using appropriate mathematical techniques.